POOJA KIRAN

Security Engineer | Application Security | AI & Agent Security | Cloud IAM

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Summary

Security engineer with an M.S. in Information Technology who finds flaws, fixes them, and tests the results. Reviewed 14 self-built security repositories; documented 1 P0 and 5 P1 findings in a dated code/CI audit. Built an MCP security gateway validated by 782 passing tests and 81.93% statement coverage in a cited Python 3.12 CI run.

Professional & Applied Experience

Independent AI Security Researcher & Engineer Aug 2024 – Present

Self-Directed Security Engineering | Tempe, AZ

- Reviewed source, tests, and CI across 14 owned security repositories; recorded 1 P0 and 5 P1 findings, remediation status, and remaining limits in a September 2026 audit.
- Corrected and regression-tested security defects involving gateway fail-closed authorization and other portfolio trust boundaries; assessed authentication, rate limits, audit trails, and workflow security.
- Used pytest, GitHub Actions, CodeQL, and dependency scans to check changes; kept fixture-only results separate from claims about real deployments.

Business & Compliance Lead | AEROSEC Aug – Dec 2025

Honeywell Aerospace × ASU | Technology Innovation Lab Externship | Tempe, AZ

- Interviewed 4 of 12 industry participants over video calls to identify vendor-access, audit-record, service-disruption, and integration risks for a proposed airline security solution.
- Coordinated work across three remote peer project leads, tracked prototype, technical and market research in virtual meetings, and communicated progress to mentors.
- Outlined API authorization, encryption, logging and vendor-risk requirements; compared four security platforms and proposed a $120,000 first-year budget with a 12-month cost plan.

Graduate Teaching Assistant | IT Grader Jan – Oct 2025

Ira A. Fulton Schools of Engineering | Arizona State University | Tempe, AZ

- Graded approximately 85 undergraduate web-programming submissions per semester and 52 graduate cybersecurity assignments against instructor rubrics.
- Explained corrections for secure coding, risk assessment, compliance, and software configuration issues.

Selected Security Engineering Projects

MCP Agent Security Gateway | Python, FastAPI, MCP/JSON-RPC, Elastic SIEM Oct 2025 – Oct 2026

- Implemented default-deny agent-to-tool authorization, anti-SSRF policies, 69 prompt-injection signatures, hash-chained audit records, and 9 Elastic detection rules.
- Reviewed HTTP/JSON-RPC, credential handling, rate limits and failure paths; cited Python 3.12 CI: 782 tests passed, 81.93% statement coverage.

AWS Agent Identity Guard | Python, AWS IAM, SARIF 2.1.0 Apr 2025 – Oct 2026

- Built 25 deterministic IAM rules for wildcard grants, iam:PassRole, sts:AssumeRole and trust-policy risks, with SARIF findings for CI review.
- Tested IAM policy analysis with 244 passing tests and 3 credential-dependent skips in a cited October 2026 CI run; no live AWS permissions claim.

Security Engineering Projects (continued)

HF Model Provenance Scanner | Python, SafeTensors, GGUF, ONNX, Keras Nov 2024 – Oct 2026

- Built non-executing checks for model serialization, embedded code, provenance and dependencies; flagged 33/33 curated adversarial fixtures.
- Tested model artifact inspection: 260 passing CI tests and 75.67% statement coverage; fixture results are not real-world detection rates.

Attack Detection Engine / **Aegis HealthSec** | Python, FastAPI, PostgreSQL, ARM/CAN Aug – Oct 2026

- Extended ATT&CK-oriented detection and offline firmware/ARM/CAN analysis with clinical-access rules and a synthetic FHIR R4 AuditEvent connector on unmerged branches.
- Built authenticated ingest-to-investigation with correlation, incident persistence and audit history; Milestone E reports 296 passing local tests, hosted CI blocked.

Technical Skills & Knowledge

Security / **risk:** Security design review, threat modeling, risk analysis and assessment, web-services security, API authorization, software testing, CI security checks

Cloud / **network** / **OS:** AWS IAM, DNS, HTTP, TLS, Zeek/Suricata-style telemetry, Linux /proc TCP, HMAC-SHA256, Ed25519

Engineering / **leadership:** Python, C++, FastAPI, Docker, pytest, GitHub Actions, CodeQL, SARIF, Elastic SIEM, remote peer coordination, compliance research, budgeting

Video protection (conceptual): DRM license flows, encrypted HLS/DASH, CloudFront signed URLs/cookies, AWS MediaPackage/SPEKE

Education & Credentials

Arizona State University | M.S., Information Technology (Security) | May 2026 | GPA 3.87/4.00

M. S. Ramaiah University of Applied Sciences | B.Tech., Computer Science & Engineering | Aug 2023

AWS Academy Graduate: Cloud Architecting (Apr 2025) | Cloud Security Foundations (Nov 2025)